

XRayVision AI — Project Documentation

Final Year Project documentation set for **XRayVision AI**, a full-stack medical diagnostic platform.

Field	Value
Project title	Web-Based AI Chatbot for Automated X-Ray Fracture Detection
Institution	Minhaj University Lahore — School of Software Engineering
Programme	BSSE, 8th Semester — Final Year Project
Students	Muhammad Ali Raza (2022F-MUL-BSSWE-017) · Hamza Afzal (2022F-MUL-BSSWE-027)
Supervisor	Maam Misbah — Lecturer, School of Software Engineering
Product version	2.1.0
Live application	https://x-ray-vision-board.vercel.app

The Documents

#	Document	What it answers	Size
1	Software Requirements Specification	<i>What must the system do?</i> Scope, users, 60+ numbered functional requirements, use case model, non-functional targets, constraints.	IEEE 830 format
2	Software Design Document	<i>How is it built?</i> Architecture, ERD, DFDs (levels 0–2), class diagram, sequence diagrams, state diagrams, algorithms, security design, deployment.	IEEE 1016 format
3	Test Documentation	<i>How was it verified?</i> Test plan, environment, test data, 93 test case specifications, traceability, defect log.	IEEE 829 format
4	User Manual	<i>How do I use it?</i> Step-by-step guide to every feature, with screenshots from the live deployment, troubleshooting and FAQ.	End-user guide
5	Thesis	The bound submission document. Title page, certificate, declaration, abstract, eight chapters, IEEE references and four appendices — drawing on documents 1–4.	FYP thesis

■ **For submission, use [XRayVision AI Thesis.docx](#)** — the thesis formatted for binding: A4, Times New Roman 12 pt, 1.5 line spacing, 1.5-inch binding margin, roman-numeral front matter, arabic body pagination, and an automatic Table of Contents field. See [Printing and Binding](#).

📄 **Every document also exists as a ready-to-open .html file in this folder** with all diagrams already rendered — no editor extension required, and it prints straight to PDF. See [Viewing the](#)

Supporting material also in the repository:

Document	Location
Enterprise Technical Documentation (27 pages, PDF)	../XRayVision AI Documentation.pdf
Project overview and quick start	../README.md
Backend setup and model reference	../backend/README.md
Security policy	../SECURITY.md
Contribution guidelines	../CONTRIBUTING.md
Code of conduct	../CODE_OF_CONDUCT.md
Database schema	../backend/supabase_schema.sql

Screenshots

All screenshots in [screenshots/](#) were captured from the **live deployment** at <https://x-ray-vision-board.vercel.app> using an automated Chromium session signed in to the demo account.

File	Page
01-landing.png	Landing page, full length
01a-landing-hero.png	Landing hero section
01b-landing-capabilities.png	Capabilities section
01c-landing-workflow.png	Workflow section
01d-landing-report.png	Report-experience section
01e-landing-trust.png	Trust/statistics section
02-login.png	Sign-in page
03-register.png	Registration page
04-forgot-password.png	Password reset page
05-dashboard.png	Dashboard with analytics
06-analyze-upload.png	Three-step analysis wizard
07-history.png	Scan history
08-chat.png	Bilingual health chatbot
09-diet.png	Diet planner

File	Page
10-clinics.png	Clinic locator
11-profile.png	User profile
12-settings.png	Settings
13-results.png	Diagnostic report with bounding boxes
14-landing-mobile.png	Landing page at 390×844

Reading Order

For the evaluation committee

1. [SRS §1–2](#) — what the system is and who it is for
2. [SDD §2](#) — the architecture
3. [User Manual §6–7](#) — the core feature in action
4. [Test Documentation §16](#) — verification status

For a developer joining the project

1. [../README.md](#) — set up and run it
2. [SDD §5](#) — where the code lives
3. [SDD §6](#) — the API contract
4. [SDD §12](#) — why it is built this way

For an end user Go straight to the [User Manual](#).

Viewing the Diagrams

The diagrams are written in **Mermaid** inside the Markdown files. Depending on how you open the docs, you may need to do nothing — or you may see nothing at all. Here is what works:

✓ Easiest — open the ready-made HTML

Every document has been pre-rendered to a **self-contained HTML file** with all diagrams baked in as SVG. **Just double-click it.** No extension, no internet connection, no build step.

Open this	For
README.html	This index
01-SRS.html	Requirements specification
02-SDD.html	Design document — all 18 diagrams rendered
03-Test-Cases.html	Test documentation
04-User-Manual.html	User manual with screenshots
05-Thesis.html	Thesis, browser edition

Or just take the PDFs

Every document is also pre-printed to A4 PDF in [pdf/](#), with page numbers in the footer. Nothing to install, nothing to render — email them, upload them, print them.

File	Pages
00-Documentation-Index.pdf	8
01-Software-Requirements-Specification.pdf	25
02-Software-Design-Document.pdf	32
03-Test-Documentation.pdf	43
04-User-Manual.pdf	27
05-Thesis-browser-edition.pdf	79

 **Which thesis file do I submit?** 05-Thesis-browser-edition.pdf is convenient for reading and emailing, but it is **not** the submission copy — it uses the web stylesheet and has no paginated contents table. **The bound copy comes from [XRayVision AI Thesis.docx](#)** : open it in Word, press **F9** on the contents field, then *File* → *Save as* → *PDF*. That version carries Times New Roman, 1.5 spacing, the binding margin and real page numbers in the Table of Contents.

You can also print any `.html` yourself with **Ctrl + P** → **Save as PDF** — the stylesheet has print rules that keep tables, diagrams and screenshots from being split across pages.

Printing and Binding

The submission copy is [XRayVision AI Thesis.docx](#) (about 11 MB, with all 22 diagrams and 13 screenshots embedded).

Formatting already applied

Setting	Value
Page size	A4
Margins	1.5 in left (binding edge), 1 in top / right / bottom
Body font	Times New Roman, 12 pt
Line spacing	1.5
Front matter pagination	Lower-case roman numerals (i, ii, iii ...)
Body pagination	Arabic, restarting at 1 from Chapter 1
Headings	Times New Roman bold, outline levels set for the contents field
Tables	27, full content width, header rows shaded and repeated
Figures	35, centred with italic captions

Before you print — two steps in Word

1. **Build the contents table.** Open the file, click the Table of Contents placeholder, then press **F9** (or right-click → *Update Field* → *Update entire table*). Word fills in the entries and their page numbers. Repeat this after any edit that changes pagination.
2. **Fill in the signature blocks** on the Certificate of Approval and Declaration pages, and sign them.

Then check against your department's template. The formatting above follows the common Pakistani university thesis convention, but departments differ on details — required certificate wording, logo placement, font choice, and whether an abstract word limit applies. Compare against the template your department issues and adjust; everything is plain Word formatting, so nothing here is locked.

To regenerate the .docx after editing. The Word file is generated from [05-Thesis.md](#), so edit the Markdown and rebuild rather than editing both. If you edit the .docx directly, treat it as the new master and stop rebuilding, or your changes will be overwritten. Build instructions are in [tools/README.md](#).

Rebuilding the Documents

The Markdown files are the single source of truth; the .html and .docx files are generated. Three scripts in [tools/](#) regenerate them:

```
cd docs/tools
pip install playwright && python -m playwright install chromium
npm install

python build_html.py          # Markdown -> HTML + diagrams/*.svg
python render_diagrams_png.py # SVG -> print-quality PNG
node build_thesis_docx.js     # Thesis Markdown -> Word
```

See [tools/README.md](#) for prerequisites, the authoring markers used in the thesis, and how to change the Word formatting.

Viewing the Markdown source instead

Where you open the .md file	Do diagrams render?
GitHub / GitLab (web)	✅ Yes, automatically
VS Code — built-in preview	❌ No. VS Code cannot render Mermaid on its own. Install the free extension Markdown Preview Mermaid Support (bierner.markdown-mermaid), then reopen the preview.
Obsidian, Typora	✅ Yes
Plain text editor	❌ You will see the diagram source code

If your diagrams are blank in VS Code, that is expected — it is the editor, not the document. Either install the extension above or just open the matching .html file.

Individual diagram files

Each diagram is also saved separately as an SVG in [diagrams/](#) — 22 files, named after their source document and position (e.g. `02-SDD-diagram-03.svg` is the ERD). Use these to paste a single diagram into a thesis chapter, a slide deck or a poster. SVG is vector, so it stays sharp at any print size.

Diagram inventory

Diagram	Type	Location
Product context	Graph	SRS §2.1
Product functions	Mindmap	SRS §2.2
Use case diagram	Graph	SRS §5.1
Analysis activity diagram	Flowchart	SRS §5.3

Diagram	Type	Location
Layered architecture	Graph	SDD §2.1
Component diagram	Graph	SDD §2.2
Entity relationship diagram	ER diagram	SDD §3.1
RLS policy map	Graph	SDD §3.5
DFD Level 0 (context)	Graph	SDD §4.1
DFD Level 1	Graph	SDD §4.2
DFD Level 2 (analysis)	Graph	SDD §4.3
Domain class diagram	Class diagram	SDD §5.3
Registration/login sequence	Sequence	SDD §7.1
Image analysis sequence	Sequence	SDD §7.2
Chat sequence	Sequence	SDD §7.3
PDF export sequence	Sequence	SDD §7.4
Scan lifecycle	State diagram	SDD §7.5
Auth session lifecycle	State diagram	SDD §7.6
Auth/authorisation flow	Graph	SDD §9.1
Degradation decision tree	Graph	SDD §10
Deployment diagram	Graph	SDD §11.1
Startup sequence	Sequence	SDD §11.2

22 diagrams across the two design documents.

Status

Document	Status
SRS	✅ Complete — 60+ functional requirements, 12 use cases, full NFR set
SDD	✅ Complete — 22 diagrams, algorithms, design rationale
Test Documentation	⚠️ Specifications complete (93 cases). 17 verified against the live deployment; 75 pending execution; 1 defect open. See the notice at the top of the document.
User Manual	✅ Complete — every feature documented with live screenshots
Thesis	✅ Complete — 8 chapters, 25 IEEE references, 4 appendices; Word edition ready for binding

Open items before final submission

1. **Update the Table of Contents field** in the `.docx` (select it and press F9) and sign the certificate and declaration pages.
 2. **Execute the pending test cases** and fill in the Actual Result columns — see [§16.1](#). Then update Chapter 6 of the thesis, which currently reports 17 of 93 executed.
 3. **Close DEF-01** — the root `.env` is tracked in git; run `git rm --cached .env` and commit.
 4. **Close DEF-02** — `backend/README.md` says the `xray-images` bucket is private; the code creates it public.
 5. **Investigate DEF-03** — the fracture detector emits non-anatomical class labels (a `vase` label was observed on a wrist radiograph). Constrain reported detector classes to the fracture vocabulary and verify the deployed weights.
 6. **Check the thesis against your department's template** — certificate wording, logo placement and font requirements vary by department.
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Medical Disclaimer

XRayVision AI is an educational tool. It is NOT a certified medical device. All outputs are AI-generated estimates intended to support, not replace, the clinical judgement of a qualified radiologist. Do not use this software for diagnosis or treatment decisions.

XRayVision AI — Project Documentation — Minhaj University Lahore